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PHYSIOLOGY OF SLEEP (REVIEW)

Interactions between stress and sleep: from basic research to clinical situations

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Acute stress is a fundamental adaptive response which enables an organism to cope with daily threatening environmental stimuli. If prolonged and uncontrollable, the stress response may become inadequate and ultimately result in health damage. Animal models of stress in rodents indicate that both acute and chronic stressors have pronounced effects on sleep architecture and circadian rhythms. One major physiological response elicited by stress is activation of the hypothalamo-pituitary-adrenal axis. In both animals and humans, the hypothalamo-pituitary-adrenal axis plays an important role in sleep–wake regulation and in alterations of the sleep–wake cycle secondary to exposure to acute or chronic stressors. In humans, dysfunction of the neuroendocrine regulation of sleep can lead to severe sleep disturbances. The progressive decay of the hypothalamo-pituitary-adrenal axis in elderly people, which mimics chronic exposure to stress, may contribute to fragmented and unstable sleep in ageing. Shift workers, chronic insomniacs or patients suffering from mental disorders show abnormal hypothalamo-pituitary-adrenal secretory activity and concomitant sleep disturbances. Those sleep disorders and possible underlying mechanisms are briefly reviewed.

Key words: stress, sleep, circadian, insomnia, depression, shift work, rhythm, PTSD, ageing, hypothalamo-pituitary-adrenal axis, catecholamines

Introduction

Walter Bradford Cannon was the first to study the various physiological responses to directly threatening environmental influences [1]. Although Cannon can be considered as the father of "stress", this term was coined for the first time by Hans Selye in 1935 [2]. Everyone has their own understanding of stress and finding a definition is difficult. Stress is a "multidimensional concept" built around at least three components: (i) the

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stimulus, or stressor, which can be positive or negative; (ii) the cognitive evaluation of the stressor, which depends on the previous life experience of the individual and on his/her ability to predict the stressful experience; (iii) the resulting physiological response(s) of the individual. This third component refers to Selye's characterization of the stress response as a "general adaptation syndrome", organized into three stages [3]. The first stage is the general alarm reaction, during which numerous biological systems (including the neuroendocrine axis) are activated in response to the stressor. The second stage would lead to resistance, with the activated biological systems returning to normal. If the stressful stimulus is maintained, the organism loses its resistance and enters a phase of exhaustion, regarded as the third stage of the syndrome (Fig. 1).

The consequences of physiological activation are many (Fig. 1): mobilization of energy (such as free fatty acids, glycerol, glucose, amino acids) from storage nutrients (triglycerides, glycogene, proteins) and ceasing further energy storage, increase in cardiovascular/pulmonary tone to facilitate tissue delivery of oxygen and glucose, slowing down of anabolic processes until the acute emergency has passed, and suppression of digestion, growth, reproduction, inflammatory responses and immunity [4]. Cognition is simultaneously altered, with a tendency towards sharpened sensory thresholds, a logical adaptation for coping with an emergency situation. At the same time, negative feedback mechanisms are activated to counteract the physiological activation and reinstate a new equilibrium. If these feedback mechanisms succeed, the organism will be able to deal with the stressful situation, eliminate its source and initiate appropriate behaviours: stress is then an adaptive response which enables the organism to cope with daily threatening environmental stimuli. If the source(s) of stress are prolonged and/or uncontrollable, feedback mechanisms fail in restoring the equilibrium; then the stress response becomes inadequate and may ultimately result in various pathological states (e.g. hypertension, cardiomyopathy, G.I. ulcerations) including sleep and mood disorders. Although many individuals experiencing stressful events do not develop such pathologies, stress seems to be a provoking factor in those individuals with particular vulnerability, determined by genetic factors or earlier experience [5].

Indirect evidence for the deleterious effects of stress on sleep comes from various studies showing sleep changes immediately after disturbing life events (e.g. death, divorce, job loss, financial loss), in chronic insomniacs [6], in shift workers [7], in stress-related disorders (such as depression [8] or post-traumatic stress disorders [9, 10]), or in elderly subjects [11]. These clinical situations will be reviewed in the second part of this article.

Through the development of various animal models of stress (see below) and human laboratory studies, progress has recently been made in the understanding of the neuroendocrine regulation of sleep, and the effects of sleep or sleep loss back on those regulating systems. This should yield new insights in the aetiology and physiopathology of human sleep and mood disorders, and lead to a better understanding of the long-term consequences of sleep deprivation.

Neuroendocrine substrates of the stress response

Major physiological responses elicited by stress are the activation of the hypothalamo-pituitary-adrenal (HPA) axis and sympatho-adrenomedullary systems. Those two systems are not separate entities, but exert mutual control on each other. Their responses are under the control of stimulatory and inhibitory inputs to the hypothalamic

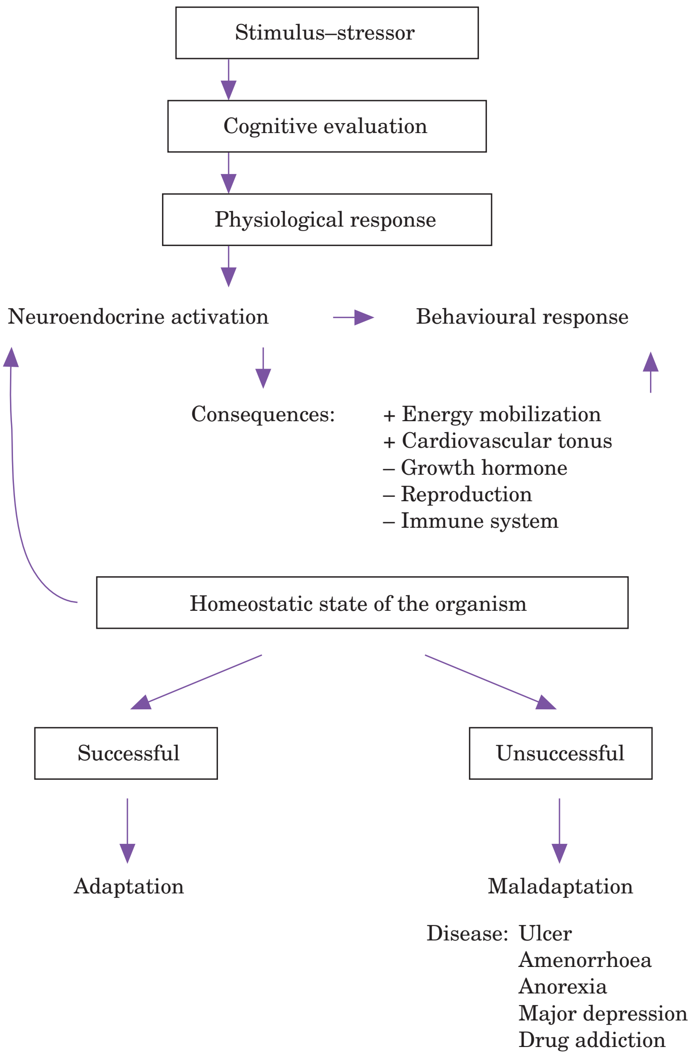

Figure 1 Schematic representation of a multidimensional concept of stress build around three important concepts: the stimulus-stressor, its cognitive evaluation by the organism and resulting physiological response(s). Activation of various biological systems (including neuroendocrine and behavioural components) leads to resistance of the organism, resulting in activated biological systems returning to normal. If the source(s) of stress are prolonged and/or uncontrollable, feedback mechanisms fail in restoring the equilibrium, the stress response becomes inadequate and may ultimately result in various pathological states, including sleep and mood disorders.

paraventricular nuclei (PVN), which control the secretion of corticotropin-releasing-hormone (CRH) and vasopressin (VP) into the pituitary portal circulation as well as other neuropeptides [12]. CRH and VP secretion leads to pituitary release of adrenocorticotropin (ACTH) and adrenal gland activation, with release of glucocorticoids. Inhibitory inputs to the PVN mainly involve the feedback action of glucocorticoids on their receptors located in the PVN and the limbic system. Corticosteroids that bind preferentially to hippocampal type I (mineralocorticoid) receptors appear to be involved in maintaining basal activity of the HPA axis, while glucocorticoid receptors mediate the effects of corticoids aimed at restoring the homeostasis in the reactive mode [12]. Those feedback mechanisms play a crucial role in the second stage of Selye's general adaptation syndrome.

The circadian timing system is another major physiological system involved in the adaptation of the organism to environmental changes. In contrast to the stress system, whose adaptive value is "reactive" to unexpected environmental changes, the circadian system has a "predictive" adaptive role, as it prepares the organism to anticipate the daily changes in the environment. Recent animal [13–15] and human [16–18] data have revealed the existence of strong interactions between the circadian system and the stress system. Those data indicate, first, that the response of an organism to an acute stressor varies as a function of the time of day of stressor presentation. They also indicate that the functioning of the circadian system is affected by stress, in such a way that its response to stress is a function of the time of day at which the stressor is applied. A better understanding of those interactions could be of high therapeutic value, especially in improving preventive strategies in some of the stress-related diseases (e.g. cardiac and neurological ischaemic diseases) whose onset demonstrate a marked circadian variation with an increased risk during the morning hours [19].

Practice Point

Reaction to unexpected environmental changes involves the stress system, with activation of the HPA axis and sympatho-adrenomedullary systems. Anticipation to daily changes in the environment involves the circadian system. A better understanding of the interactions between those two adaptive systems may have important therapeutic implications

Sleep and circadian rhythms in animal models of stress

Both acute and chronic stressors have pronounced effects on sleep architecture and circadian rhythmicity in rodents. Recovery from acute stress usually involves a reparative sleep rebound, characterized by an increase deep in slow wave sleep (SWS) and in rapid eye movement (REM) sleep. In rats, for instance, an acute "immobilization stress" procedure decreases SWS and suppresses REM sleep as long as the animal is immobilized. After release, there is a significant increase in REM sleep and SWS [20, 21], and altered circadian rhythm of corticosterone secretion [22]. "Social defeat" in rats, a natural stressor causing strong neuroendocrine and behavioural stress responses, also induces changes in sleep and circadian rhythmicity. The major sleep change consists of a sharp increase in EEG slow-wave activity (SWA) during deep SWS. SWA has been identified as an indicator of sleep intensity [23], which suggests that acute stressors may accelerate the build-up of a sleep debt [24]. In hamsters, acute induction of arousal/locomotor activity at a time of usual inactivity/sleep, or acute immobilization at a time of usual intense activity, are both powerful stressful stimuli for the animal and behavioural resetting stimuli for the circadian clock [13, 14]. These data emphasize the interactions between "stress-based" and "circadian time-based" adaptive physiological systems.

In rats, hypothalamic CRH has been implicated in the regulation of physiological waking [25] and in sleep-wake modifications induced by acute stress exposure, primarily manifested by changes in REM sleep [21]. Stress-induced elevation of plasma ACTH is associated with an increase in REM sleep and to a lesser extent in SWS sleep [26]. Effects of acute stress on sleep seem primarily to involve central mechanisms (rather than the HPA axis), since adrenalectomy does not modify those effects [27]. Both the noradrenergic and serotonergic systems are thought to play a permissive role in REM sleep regulation [28]. Under stress conditions, CRH acting as a neurotransmitter in the locus coeruleus, induces an increase in activity of noradrenergic neurons, which leads to an increase in REM sleep [21, 29]. Serotonergic neurons in the raphe, activated by acute immobilization stress, are important in the mediation of the restraint stress-induced ACTH response [30].

Various paradigms involving chronic stress can also affect sleep architecture in rodents, mainly through a sleep-disrupting effect [27]. For example, rats exposed to chronic “intermittent foot shock” or “learned helplessness” exhibit an increase in REM sleep during the first day of recovery [31]. Chronic “unpredictable mild stress” in rats induces a decrease in latency to REM sleep onset, an increase in REM sleep and a decrease in deep SWS [32]. In prolonged exposure to immobilization stress, reparative SWS and REM sleep rebounds are eliminated [27]. Interestingly, the immobilization-induced corticosterone secretion appears to play a role in this suppression of sleep rebound, since adrenalectomized rats show an increased sleep rebound after immobilization compared to controls [27], and treatment with the glucocorticoid receptor agonist, dexamethasone, can mimic the effects of prolonged immobilization stress on sleep [27]. Exposure of rats to chronic stressors also alters their behavioural, endocrine and autonomic circadian rhythms [33]. Similarly, manipulations used to induce sleep deprivation (total [34] or selective REM [35] sleep) in rats activate the HPA axis, indicating the stressful nature of the methods used to induce sleep deprivation in rodents.

Interestingly, the effects of chronic stressors on sleep seem dependent on the nature, the timing and the duration of applied stressors. For instance, all the above-described stress-induced changes in sleep and circadian rhythms in rats are transient and disappear in the days following the end of the stressor. In contrast, exposure of rats to stress *in utero* can lead to profound endocrine [15] and behavioural [36] abnormalities that are persistent and stable over time [36].

Pre-natally-stressed (PNS) rats tested during adulthood exhibit abnormally prolonged stress-induced corticosterone secretion [37]. Under baseline conditions, those rats show an increase in REM sleep and light SWS, higher sleep fragmentation, and a slight decrease in deep SWS (Fig. 2). During recovery sleep from acute immobilization stress, all those baseline sleep changes persist and are correlated with abnormal stress-induced corticosterone secretion in PNS rats [36]. Those PNS rats also show disturbances in circadian rhythms (locomotor activity, corticosterone secretion) that are consistent with those observed in depressed patients [15, 38]. Those sleep abnormalities in rats are consistent with reports of abnormal “sleep-like behaviours” in pre-natally-stressed humans [39].

We have recently observed sleep abnormalities in Wistar Kyoto (WKY) rats, a strain of rats with depressive and anxiety-like behaviours, abnormal circadian rhythms of corticosterone and thyroid-stimulating hormones and hyper-response to stress [40]. Under baseline conditions, WKY rats show a 50% increase in REM sleep during the light phase and an increase in sleep fragmentation during both the light and dark phase. After a 6-h sleep deprivation, REM sleep rebound is more pronounced during

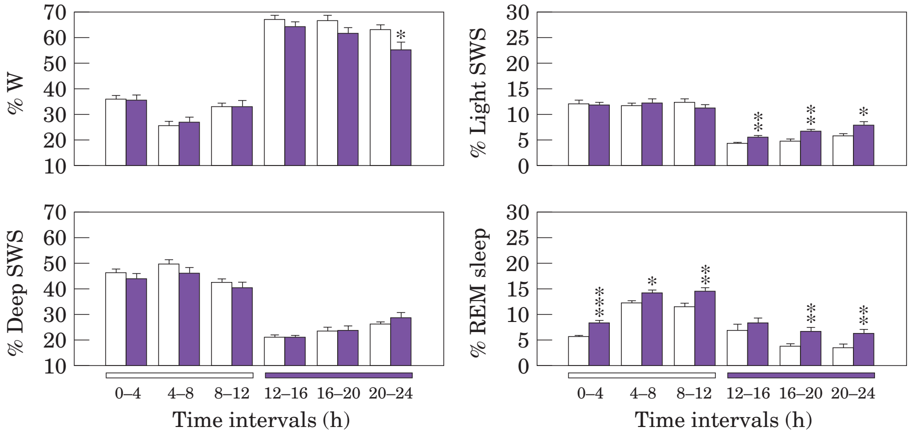

Figure 2 Distribution per 4-h intervals of vigilance states in eight control (CONT: □) and eight pre-natally-stressed (PNS: ■) rats under baseline conditions. Animals were subjected to a 12:12 light–dark cycle (lights on, 00:00–12:00 h; lights off, 12:00–24:00 h). Mean (±SEM) values of wake (W), light slow-wave sleep (SWS1), deep slow-wave sleep (SWS2) and REM sleep are expressed as percentage of recording time. PNS rats showed an increased time in REM, during both the light and dark phases. Those changes result from an increase in the number of REM episodes, while their mean duration remained the same. Pre-natal stress also induced an increase in total SWS1 time, restricted to the dark phase. * $P<0.05$, ** $P<0.01$, *** $P<0.001$ (two-tailed unpaired Student's t-test) for between-groups comparisons (adapted from [36]).

the dark phase in WKY rats. As the WKY rat represents a genetic model for depression with altered EEG sleep patterns and abnormal responses to stress, this strain may be particularly useful for investigating the relationship between depression, stress and sleep abnormalities.

Practice Point

Both acute and chronic stressors have pronounced effects on sleep architecture in rodents. Recovery from acute stress usually involves a reparative sleep rebound characterized by an increase in REM sleep and SWS, driven by central mechanisms. Exposure to chronic stress leads to fragmented sleep, probably driven by a stress-induced increase in corticosteroids.

Mechanisms involved in the interactions between stress and human sleep

In humans, there is a close and robust temporal association between sleep structure and HPA axis activity. The early phase of nocturnal sleep, dominated by extended epochs of SWS, is the only time of day during which secretory activity of the HPA axis is subject to a pronounced and persistent inhibition, resulting in minimum concentrations of ACTH and cortisol, with concomitant high levels of growth hormone

(GH) [41]. In contrast, during late sleep, predominated by REM sleep, HPA secretory activity increases to reach a diurnal maximum shortly after sleep awakening. Also, while arousals during light SWS are associated with bursts of sympathetic-nerve activity, during deep SWS sympathetic-nerve activity is reduced compared to wakefulness [42]. Activities of the HPA axis and sympathetic system are positively correlated to the overall amount of REM sleep [43]. The implication of the daily fluctuations of ACTH in sleep regulation has been recently outlined in a study showing that a morning rise in ACTH seems to play a critical role in timing the end of nocturnal sleep [44].

On the other hand, sleep–wake transitions (i.e. from wakefulness to sleep or from sleep to wakefulness) affects the functioning of the HPA axis. Sleep onset is reliably associated with decreasing or low cortisol levels [45, 46]. This might be related either to the high amount of SWS during the first hours of sleep [47], or to preparatory mechanisms facilitating sleep onset since cortisol secretion has been shown to have an inverse relationship to SWA, with cortisol secretions preceding SWA variations by about 10 min [45, 48]. Conversely, awakenings during or at the end of the sleep period are consistently followed by a pulse of cortisol [49]. The recent demonstration of temporal coupling between cortisol secretion and EEG β activity (i.e. an index of central alertness) during wakefulness is consistent with this observation [50].

Exogenous administration of each of the major mediators of the HPA axis (i.e. CRH, ACTH and cortisol) has shown effects on sleep architecture. Pulsatile administration of CRH produces several hormonal and sleep changes (reduced SWS, reduced REM sleep during the second part of the night) that resemble those found in depression [51]. Intravenous administration of ACTH was found to delay sleep onset, reduce SWS and induce fragmented sleep [52]. Single dosed, continuous or pulsatile cortisol infusions increase SWS and decrease REM sleep [51]. The effects of different corticoids on sleep may involve the activation of different corticosteroid receptor types. Activation of mineralocorticoid receptors increases the time spent in NREM sleep, while binding to glucocorticoid receptors increases time spent awake or in REM sleep [41].

In considering possible mechanisms underlying the interactions between stress and sleep, one cannot ignore the role of the immune system and its response to stress [53]. Acute or chronic stressors have a strong impact on immune function in both animals and humans. *Acute* stressors have an activating impact on immune function which concerns mainly the number of natural killer (NK) cells and may be mediated by catecholamines [54]. In contrast, *chronic* stress can lead to a downregulation of the immune response, with fewer B and T lymphocytes cells and decreased natural killer cell activity [55]. A number of stressful life events have been associated with decreases in immune function, such as bereavement [56], divorce or marital difficulties [57]. Similarly, affective states associated with stress, such as depression [58] or post-traumatic stress disorder [59], have been associated with reduced immune defences. Various studies on the role of cytokines have pointed out the important roles of interleukin-1β (IL-1β), tumour necrosis factor (TNF) and interferon in sleep regulation [60]. Administration of exogenous IL-1β or TNF induces increased NREM sleep, while their inhibition reduces spontaneous sleep [61]. IL-1β also participates in an immunoregulatory feedback loop leading to the activation of the HPA axis; this represents one of the links between stress and sleep regulation. Available data in humans show that plasma IL-1β levels vary in phase with the sleep-wake cycle, and plasma TNF levels in phase with EEG slow-wave activity, and that both IL-1β and TNF increase during sleep deprivation [62]. Also, strong associations between disruptions of sleep continuity and impaired NK cell function have been reported [58].

Practice Point

In humans, there is a close and robust temporal association between sleep–wake states and activity of the HPA axis and the sympathetic system. Exogenous administration of each of the major mediators of the HPA axis has effects on sleep architecture. Various components of the immune system also have an impact on sleep structure. Immune function, which in turn can be altered by acute or chronic stressors, may thus play an important role in mediating some of the interactions between stress and sleep

Sleep deprivation and stress

An important step in responding to a stressor is the subjective evaluation of the stress by the subject. The perception of short-term sleep deprivation does not correspond to the usual perception of an acute stress. Indeed, in human laboratory studies, during and after a night of total sleep deprivation, subjects do not report being more "tense" or less "calm" on visual analog scales (Leproult *et al.*, unpublished data), suggesting that in these experimental conditions sleep deprivation is not perceived as a stressor. This could be due to the fact that in such studies subjects are aware of the modification and the experimental procedure. They know the exact duration of the sleep deprivation and that they will be able to recover their sleep need immediately after the study.

However, there are arguments to consider sleep deprivation as a stressor. One comes from the immediate effects of sleep loss on the HPA axis at the usual time of sleep onset. Since the modulatory effects of sleep–wake transitions on cortisol release are absent under sleep deprivation, the cortisol profile shows mild alterations resulting in reduced amplitude. The absence of sleep onset is indeed associated with higher cortisol levels at the quiescent period of the rhythm [63]. A more robust argument in favour of the stressful nature of sleep deprivation arises when we consider its effects on the HPA axis on the following day. Under regular sleep–wake conditions, the 24-h profile of plasma cortisol in young adults show an early morning maximum and declining levels throughout the day, with a quiescent period centred around midnight and a rapid elevation in late sleep. In contrast, a state of sleep loss induced in healthy young men either by partial or total sleep deprivation or by semi-chronic curtailment of the sleep period, results in an elevation of evening cortisol levels the following day and in a delayed onset of the quiescent period of cortisol secretion (Fig. 3) [64, 65]. The normal day-long decline of cortisol levels partially reflects the recovery of the HPA axis from the early morning circadian stimulation that occurs in response to increased CRH drive during the second part of the night. Elevation of evening cortisol levels might thus reflect an alteration of the rate of recovery of the HPA axis from this endogenous challenge which is likely to involve an impairment of the feedback regulation of the HPA axis mediated by the hippocampus.

Since sleep loss seems to affect this HPA resiliency in young subjects, the ability of the HPA axis to recover from exogenous stimulation by stressors may also be affected by sleep loss. The superimposed effect of stress and sleep loss experienced in a chronic manner is likely to be associated with long-term exposure to cortisol, yet prolonged exposure to cortisol increases the natural vulnerability of neurons and is supposed to

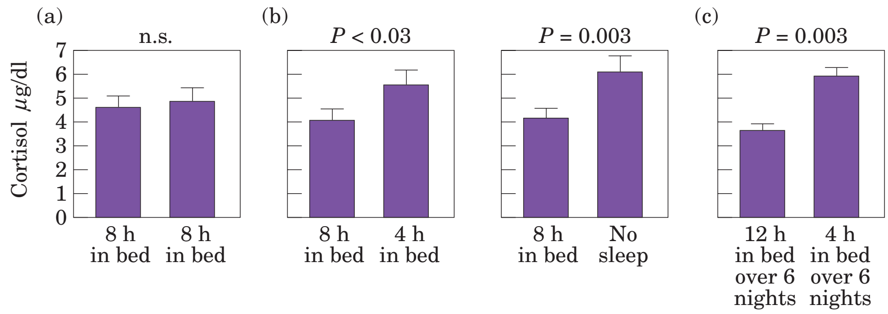

Figure 3 Evening cortisol levels observed in three groups of young healthy subjects before and after they were studied in three different conditions. (a) A normal bedtime condition in which sleep was allowed from 23:00 to 07:00 h. (b) An acute sleep loss condition, either partial with a 4-h bedtime period from 04:00 to 08:00 h (left part of the middle panel) or total (right part of the middle panel). Acute sleep loss was associated with 30–50% higher cortisol levels in the later part of the day. (c) A semi-chronic sleep loss condition during which subjects were allowed only 4 h in bed for 6 days. This condition was also associated with higher cortisol levels in the later part of the day. Cortisol levels were measured in each subject on two separate occasions: under baseline conditions (left column in each bar graph) and after sleep manipulation (right column in each bar graph) (adapted from [64, 65]).

accelerate hippocampal deterioration [66]. As the hippocampus is the main structure involved in the feed back regulation of cortisol release, a situation of stress and chronic sleep loss would further promote alterations in the feedback mechanisms involved in the control of the HPA axis. Chronic sleep loss may therefore accelerate the development of metabolic and cognitive consequences of glucocorticoid excess, such as cognitive deficits and decreased carbohydrate tolerance [67, 68]. Recent studies indicate that populations of developed countries chronically sleep deprive themselves because of their socioeconomic and cultural environments. In addition to the adverse consequences on mood and alertness, the possible central and peripheral disturbances associated with glucocorticoid excess in chronic sleep loss may lead to long-term adverse health effects.

Such an elevation of evening cortisol levels also occurs in a variety of conditions including depression [69] and ageing [70]. It is possible that the sleep fragmentation which characterizes both depression and ageing play a role in this abnormality of HPA function. Interestingly, the functional component of HPA regulation that is most affected by ageing is the rate of recovery from a challenge, i.e. the resiliency of the response, rather than the unstimulated levels or the magnitude of the response [71].

Practice Point

Experimental evidence is in favour of the stressful nature of sleep deprivation in humans. Chronic sleep loss may therefore accelerate the development of metabolic and cognitive consequences of glucocorticoid excess, such as cognitive deficits and decreased carbohydrate tolerance.

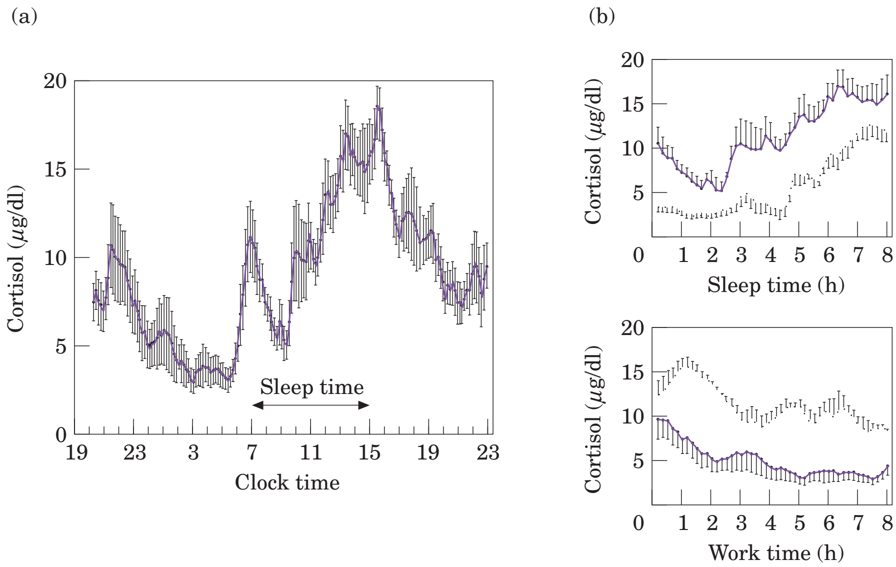

Figure 4 (a) 24-h cortisol rhythm in permanent night workers. Their sleep schedule (07:00-15:00 h) is shown at the bottom of the panel. There is a clear distortion of the cortisol rhythm: the acrophase shows good adaptation, whereas the quiescent period, abruptly interrupted by a large peak, underwent only a partial shift of about 3 h, leading to its dissociation from the sleep episode. (b) Plasma cortisol levels during the usual sleep period (top) and the usual work period (bottom) in night workers (——) compared to day-active workers (– – –). In night workers, the sleep period coincides with high cortisol levels, whereas the work period coincides with the quiescent period of cortisol secretion (adapted from [73, 112]).

Shift work and stress

More and more businesses provide their full range of services 24 h per day, 7 days a week. To meet these demanding challenges, more than 20% of the working force in industrialized countries work permanently at night, or on schedules requiring a rotation of day, evening and night work. Around the clock operations are associated with major changes in light–dark cycles and the behavioural control of sleep–wake cycles, leading to desynchrony between circadian rhythms and environmental periodicities. Workers on permanent or rotating night shifts do not adapt to these schedules, even after several years on the job [72, 111]. Figure 4 shows that the temporal profile of cortisol shows only partial adaptation to the work schedule [73]. Therefore, workers are required to work during the “wrong” phase of their circadian rhythms, when they are mostly sleepy, inefficient and prone to accidents and to try to sleep at an unadapted circadian time, leading to chronic behavioural and physiological maladaptation.

Shift work is a major health hazard, involving an increased risk of stress-related diseases: ischaemic heart disease, gastrointestinal disorders, psychosocial complaints, substance abuse, sleep disturbances, reduced immune function and infertility (see [74] for review). Sleep problems are certainly the most sensitive index of health dysfunction in shift workers [75]. About 60–70% of shift workers complain of their sleep: most of them report difficulties falling and staying asleep, poor quality of sleep, and difficulties staying awake at work. Dislocation of circadian rhythms affect most shift workers, but only 20–30% of them tolerate it poorly, suggesting that additional coping factors might be important. Indeed, many of the unspecific complaints of night workers, such as fatigue, malaise, difficulty concentrating and irritability, can result from sleep loss [76]. Sleep after night work is usually shorter and more fragmented than sleep after day work [7]. Shortening of daytime sleep after a night shift mainly affects REM sleep and stage 2 of NREM sleep. Reduced effects on SWS were attributed to the sleep pressure due to the extended duration of prior wakefulness. In a recent study investigating daysleep quality in night shift nurses (alternating between day and evening schedules) a reduction in SWS was reported, whether or not the nurses had subjective sleep complaints [77]. Sleep during the daytime may be altered by both endogenous and exogenous factors. Endogenous factors include high and rising body temperature, high circulating levels of hormones associated with wakefulness (such as cortisol, see Fig. 4), and low levels of hormones associated with sleepiness (such as melatonin). Exogenous factors include, among others, stress, traffic, children playing and household noise.

Over time, various models have been proposed to explain the relationship between shift work and health. Those models have evolved from a linear relationship of shift work, circadian disturbances and health to modelling a dynamic relationship characterized by multidirectional pathways, incorporating the important roles of psychological variables such as coping and cognitive appraisal (for review see [78]). The stress component is thought to play a major role in linking shift work and health consequences. Stress can be viewed in terms of "circadian rhythm disturbance"; stress is then the temporal disorder of physiological performance, caused by the discrepancy between the time structure of behaviour and the normal phases of circadian rhythms [78]. However, characteristics of shifts (i.e. length, direction, rotation speed), biological disturbances and stable individual differences are not enough to explain fully individual variations in adaptation to shift work. Indeed, shift work is only one of several job stressors (which include lack of autonomy, time pressures, monotony) encountered by those workers. In addition they are also exposed to non-occupational stressors. Indeed, part of the stress of shift work is due to social factors, such as frequent absence from the family with disruption of its organization, insufficient recreation, parenting and the sexual role, resulting in family tension and pre-occupation.

Practice Point

The stress component is thought to play a major role in linking shift work and health consequences, including sleep disturbances. Sleep in shift workers can be altered by both endogenous (i.e. circadian, metabolic, hormonal) and exogenous (traffic, children playing, household noise) stress-related factors. Social stress factors (disruption of family organization, parenting role) also contribute to sleep fragility in shift workers.

Sleep in selected stress-related disorders

Dysfunction of the above described neuroendocrine regulation of sleep can lead to severe sleep disturbances. For instance, elderly subjects [11] or depressed patients [8] all show insufficient inhibition of HPA secretory activity (particularly prominent during early sleep) and concomitant reduced SWS. Patients with chronic fatigue syndrome

[79] or fibromyalgia [80] show features of secondary adrenocortical deficiency of central origin. The "burnout" syndrome is characterized by lower cortisol secretion and higher cortisol suppression with dexamethasone [81]. A recent study in chronic insomniacs without major psychiatric disorder shows that activity of both limbs of the stress system (i.e. the HPA axis and the sympathetic system) relates positively to the degree of sleep disturbances, as objectively measured by polysomnographic indices [6]. These data in chronic insomniacs fit with results of previous studies showing a higher urinary excretion of 17-hydroxysteroids in poor sleepers than in good sleepers [82]. Selected clinical situations characterized by stress-related sleep disturbances will be briefly described.

Ageing

The greater prevalence of disturbed sleep with increasing age has been found both in surveys and in laboratory studies. In contrast to young adults, the elderly show a reduction of SWS and a decrease in GH secretion at the beginning of the night, a shortened REM latency and a flattened night-time rise of cortisol secretion, which also occurs prematurely during the night [70]. Additional sleep characteristics in elderly subjects include shortening of total sleep time and increased number of nocturnal awakenings. Insomnia in the elderly is often the product of multiple factors that can be difficult to separate [83].

The decay of the HPA axis may contribute to fragmented and unstable sleep in the elderly. Diurnal rhythmicity of cortisol secretion is preserved in old age, but its relative amplitude is dampened, and timing of the circadian elevation is advanced [84]. Also, the level of the nocturnal nadir of cortisol increases progressively with age. Those changes in the levels and diurnal variation of adrenocorticotropic activity are consistent with the hypothesis of the "wear and tear" of lifelong exposure to stress [4]. Cumulative exposure to glucocorticoids causes hippocampal defects, resulting in an impairment of the ability to terminate glucocorticoid secretion at the end of stress and, therefore, in increased exposure to glucocorticoids, which in turn further decreases the ability of the HPA axis to recover from stress. Such changes in circadian amplitude and phase could be involved in the aetiology of sleep disorders in the elderly [84].

Many other factors are contributory to disturbed sleep in older people, possibly through HPA disturbances, e.g. sleep disordered breathing, periodic leg movement, medication, physical illness, mood disorder, socioeconomic problems and life stress. Even in the healthy elderly, those experiencing the death of a spouse are much more likely to have complaints of chronic insomnia. Indeed, late life sleep disturbances have been associated with specific stressful events such as bereavement [85], retirement, or divorce [86]. In those cases, the insomnia is likely to be due to both the functional (change in lifestyle) and the emotional consequences of those events. Interestingly, good and poor elderly sleepers have different susceptibilities to similar stressful life events, suggesting that ongoing life stress may play a role in the perpetuation of chronic insomnia in older poor sleepers [11].

Adjustment sleep disorder

Also called transient psychophysiological insomnia or acute stress insomnia, this disorder represents sleep disturbance temporally related to acute stress, conflict and environmental changes causing emotional arousal [87]. Symptoms usually develop in association with the identified stressor (death of a loved one, divorce, financial problems, exams, unfamiliar sleep environment) and revert if the stressor is removed or the level of adaptation is increased. Such patients may experience interference with sleep onset and early morning awakenings.

Depression

Almost every depressed patient complains of insomnia. Recent studies also show that chronic insomnia is indicative of a greater long-term risk for subsequent clinical depression and psychiatric distress [88]. Subjective complaints of depressed patients include prolonged latency to sleep, frequent and prolonged awakenings at night and early morning awakening, vivid dreams, unrefreshed sleep and daytime tiredness [89]. Objective polysomnographic measurements in depressive patients show decreased SWS, reduced REM sleep latency, increased REM sleep amount and density during the first sleep cycle [90].

Stress is often a predisposing factor in the development of insomnia [91] and depression [92, 93]. Studies aimed at establishing a causal relationship between stress, hypercortisolism and vulnerability to depression have often pointed out to previous or early life experience, both in animals (see section on animal models of stress) and humans. Vulnerability factors named in studies documenting neuroendocrine abnormalities and depressive states include traumatic life events, work or marital problems, maternal separation or childhood abuse. Hyperactivity of the sympathetic nervous system and the HPA axis with hypercortisolism are seen in about 50% of depressed patients, and can be reversed by successful antidepressant therapy [94, 95]. In these depressed patients, CRH and VP expression in the PVN is enhanced, adrenals show hypertrophy and basal corticosteroid levels are increased [96]. When challenged with cortisol, dexamethasone, CRH or the combined dexamethasone-CRH test, those patients show feedback resistance at the level of the PVN and pituitary [12]. Current hypotheses hold that mechanisms underlying abnormal HPA function are causal factors in the development and course of depression. Studies in depressed patients have shown that HPA axis abnormalities are positively correlated with amounts of night-time waking or light SWS during sleep, and negatively with amounts of deep SWS [97]. After adequate treatment of depression, normalization of initially disturbed HPA axis indicates a good prognosis, and persistent HPA dysregulation is associated with a greater likelihood of relapse or chronicity [98].

It has been proposed that HPA hyperactivity in depression is probably initiated and/or maintained by the combination of enhanced central production of CRH and desensitization of glucocorticoid receptor binding system in the hippocampus, the central regulator of HPA activity. Evidence has emerged that corticosteroid receptor function is impaired in many depressed patients and in healthy individuals at increased genetic risk for a depressive disorder [99]. If this is true, then secretion of CRH would be enhanced in various brain areas, accounting for a variety of depressive symptoms. Feedback resistance and basal hypercortisolism, which characterize depressed patients, are already present in healthy subjects at high risk for affective disorders [99]. Those individuals already have an imbalance in drive and feedback inhibition of their HPA axis which precedes clinical manifestations and becomes harmful only during conditions of chronic exposure to stress.

As shown in animals with impaired glucocorticoid receptor function, antidepressants enhance the signalling through corticosteroid receptors [100]. This mechanism of action can be amplified through blocking central mechanisms that drive the HPA system, suggesting that depressive patients with HPA hyperactivity may profit from treatments with agents blocking excessive corticoid signalling, such as metyrapone [101], corticosteroid antagonists [102], or the promising CRH receptor antagonists [103, 104].

Recent work has revealed a putative important role for neuroactive steroids in the maintenance of homeostasis during hormonal response to stress by counteracting the activity of the HPA axis [105]. Interestingly, some of those compounds have an agonistic modulator effect at the level of GABAa receptors. Fluctuation in the concentrations of neuroactive steroids could in part be responsible for the increased vulnerability to develop certain psychiatric diseases. Moreover, rats receiving the antidepressant fluoxetine show an increase in brain synthesis and content of neuroactive steroids (e.g. allopregnanolone), suggesting a possible anxiolytic and antidysphoric actions of this drug via neural steroids [106]. Indeed, the imbalance of these steroids in patients can be corrected by treatment with antidepressant drugs [107].

Post-traumatic stress disorder

Victims of disasters or people trying to adapt to traumatic events typically experience recurring and distressing thoughts about the stressful event, and attempt to avoid behaviours associated with the event. Those intrusive and uncontrollable images may play a role in maintaining the stress and its deleterious effects long after the stressor is gone. Sleep disturbances are cardinal symptoms in patients suffering from post-traumatic stress disorder (PTSD). Fragmented sleep caused by frequent nightmares, re-experiencing the trauma, is the primary sleep disturbance in such patients. The frequency of nightmares has a strong relationship to the level of exposure to the traumatic event [108]. To a lesser extent, those patients experience sleep onset and/or maintenance insomnia. Polysomnographic findings in PTSD patients vary from one study to another but usually include reduced sleep efficiency, longer time awake, elevated awakening and movement time and paradoxical elevated waking thresholds during sleep [10, 109]. Many patients with PTSD often meet clinical criteria for major depression. However, HPA axis sensitivity in PTSD is different from what is found in depression. PTSD is characterized by enhanced negative feedback of HPA axis, which results in a lowered setpoint of HPA activity, with higher CRH levels [17] and reduced daily production of cortisol [110]. Differences in sleep characteristics and HPA abnormalities between PTSD and depression may help differentiating the conditions.

Practice Point

Severe sleep disturbances associated with dysfunctions of neuroendocrine regulation are observed in elderly subjects and in patients suffering from various mental or physical disorders, such as depression, chronic fatigue, fibromyalgia, PTSD, burnout syndrome or various forms of insomnia.

Conclusions

The HPA axis, central catecholamine systems and sympathetic system play an important role in the regulation of the sleep–wake cycle. Research findings also suggest that circadian rhythms, sleep and wakefulness states can influence the functioning of those regulatory systems. The set point of the HPA axis is genetically determined but can be modulated and reset to other levels by early or later stressful experiences. In animal models of stress and in various clinical situations, there is good evidence that dysregulation of the neural and neuroendocrine mediators of the stress response can lead to sleep and/or mood disturbances. Depending on the type of dysregulation and the individual's genetic factors or earlier experience, vulnerability may develop to different sleep or mood disorders.

Research Agenda

Further development of animal models involving early life environmental manipulations or genetic modifications (such as inbred strains and mutations) responsible for differences in sleep phenotypes should allow the testing of various hypotheses regarding the importance of physiological state or genetic background in the regulation of sleep and its disturbances. These strategies will help in clarifying the question of individual differences in the impact of stress on sleep, and elucidate the mechanisms that link sleep, the circadian clock and stress systems to each other and to mood alterations. This approach will lead to new experimental paradigms in humans aimed at a better characterization of sleep disorders, and the relative contribution of genetic and environmental factors in their symptomatology. It should open the way to the development of more specific therapeutic tools in sleep medicine, leading to a better integration between pharmacological and non-pharmacological approaches in the treatment of sleep disorders.

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